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Program: Data Science and Machine Learning Summer Internship

GitHub Repository Link:

Project Report: Quality Prediction in a Mining Process

1. Background & Introduction

Industrial flotation plants play a critical role in the beneficiation of iron ore by separating valuable minerals from waste gangue materials. In this process, continuously monitoring and controlling the quality of the final product—specifically the **percentage of silica concentrate**—is vital for downstream steel manufacturing efficiency. Traditionally, silica concentration levels are measured via laboratory tests, which introduce significant time delays. Developing a machine learning regression model to predict silica concentrate levels in real time using high-frequency sensor telemetry allows plant engineers to anticipate quality fluctuations and make proactive operational adjustments.

2. Problem Statement

The primary objective of this project is to build a robust and accurate regression model using multi-rate time-series sensor data from an industrial flotation plant to predict % Silica Concentrate.

A key technical challenge in this dataset is avoiding **data leakage**. Because % Iron Concentrate and % Silica Concentrate share an inverse mathematical dependency and mass balance constraints within the process, including iron concentration as a feature artificially inflates model accuracy while invalidating its real-world predictive utility. Therefore, the model must forecast silica quality strictly using operational variables (such as air flows, pulp pH, densities, and reagent flow rates) *without* relying on iron concentration.

3. System Design & Methodology

The data pipeline was structured into distinct, reproducible phases within a Google Colab environment:

- **Data Ingestion & Cleaning:** Handled large-scale industrial CSV logs containing over 730,000 records, converting European-style comma-separated decimals into standard floating-point values and handling missing telemetry data.
- **Feature Engineering:** Extracted temporal components (year, month, day, hour) from plant timestamps to capture cyclical operational trends.

- **Data Leakage Prevention:** Explicitly dropped % Iron Concentrate from the feature matrix (X).
 - **Model Training:** Utilized an **XGBoost Regressor** optimized with gradient boosting trees to capture complex non-linear relationships across high-dimensional industrial sensors.
 - **Evaluation & Persistence:** Split data into an 80-20 train-test partition and saved the production artifact as Mining_Quality_Model.pkl.
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4. Implementation Details

The core code workflow executed in the Jupyter Notebook (Project10_Quality_Prediction.ipynb) included:

Python

```
import pandas as pd
```

```
import numpy as np
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
from xgboost import XGBRegressor
```

```
import joblib
```

```
# 1. Feature and Target Separation (excluding data leakage)
```

```
target = "% Silica Concentrate"
```

```
drop_columns = ["% Iron Concentrate"]
```

```
X = df.drop(columns=drop_columns + [target])
```

```
y = df[target]
```

```
# 2. Train-Test Split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)
```

```
# 3. Model Training
```

```
model = XGBRegressor(n_estimators=300, learning_rate=0.05, max_depth=6,
random_state=42)

model.fit(X_train, y_train)
```

4. Persistence

```
joblib.dump(model, "Mining_Quality_Model.pkl")
```

5. Results & Performance Metrics

The trained XGBoost regression model was evaluated on the unseen test dataset ($N = 147,491$), achieving strong predictive performance:

- **Mean Absolute Error (MAE):** 0.4129
- **Root Mean Squared Error (RMSE):** 0.5614
- **R^2 Score:** 0.7506

An analysis of feature importance indicated that reagent flow rates (such as amine and starch flows) and pulp chemistry variables carry the highest predictive weight in determining final silica impurity levels.

6. Conclusion & Learnings

This project successfully demonstrated the viability of applying gradient boosting regression to industrial time-series data. Key takeaways include:

- Successfully identifying and neutralizing data leakage vectors (excluding iron concentration).
- Managing large-scale industrial telemetry datasets with custom string-to-float cleaning pipelines.
- Deploying an interpretable machine learning model capable of providing early quality warnings to plant operators.